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Sub ject : DSA

Unit 2, S1, SLO2 - Inserting an Element into an Array

Algorithm to append a new element VAL to an existing array

Step 1 : Set upper_bound = _____

Step 2 : Set A[_____] = _____

Step 3 : EXIT

Answer: Step 1: upper_bound + 1

Step 2: A[upper_bound] = VAL

Step 3: EXIT

Algorithm to insert an element VAL in the middle of an array given a position POS

Step 1 : [INITIALIZATION] SET I=N

Step 2 : Repeat Steps 3 and 4 while I >= _____

Step 3 : SET A[_____] = A[____]

Step 4 : SET I = _____

[END OF LOOP]

Step 5 : SET N = _____

Step 6 : SET A[POS] = _____

Step 7 : EXIT

Answer: Step 2: while I >= POS

Step 3: SET A[I + 1] = A[I]

Step 4: SET I = I - 1

Step 5: SET N = N + 1

Step 6: SET A[POS] = VAL

Step 7: EXIT